Mitigating Smart-Charging Rebound Peaks Under Network-Aware Dynamic Pricing on Low-Voltage Feeders

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***Abstract*—The rapid adoption of electric vehicles (EVs) challenges low-voltage networks. A home charger draws as much power as several households, so many vehicles charging at once can overload the street's feeder. Time-of-use tariffs move charging away from the evening peak, but every driver sees the same cheap period, creating a rebound peak when it begins. Network-aware tariffs avoid this by pricing each half-hour from the feeder's spare capacity, assuming drivers respond differently. Smart chargers do not: they run similar algorithms on the same price, so they all choose the same cheapest half-hours. This research models such chargers, including a commercial scheduler, on a 75-home London feeder using Monte Carlo simulation, and tests three mitigations: a random start delay, a household-specific price offset (a dither) and a cap on EV power per half-hour (a quota). At full EV ownership, every shared price raised the 95% conditional value at risk of the daily peak from 129 kW to 147–151 kW. The delay left 36% of days over the 100.9 kW limit, a ±2 p/kWh dither 7.0% and a quota 5.5%. The dither suffices up to 80% EV ownership; beyond that, a quota is needed.**

***Index Terms*—Demand response, distribution networks, dynamic pricing, electric vehicle charging, rebound effect, smart charging.**

I. Introduction

The homes on a street are supplied by one low-voltage (LV) cable, called a feeder, whose limit is set for ordinary household demand. On the London feeder studied here, 75 homes draw about 1 kW each at the evening peak, against a limit of 100.9 kW [1]. A home EV charger draws 7.4 kW, as much as about seven homes, so the spare capacity is used up by only three cars. When more cars than that charge at once, demand breaches the limit, and the cables and transformer overheat or trip [1], [2].

Fig. 1(a) shows that the problem is the evening: drivers plug in when they get home, just as household demand peaks. Overnight, the same feeder has room for about ten chargers, and most cars stay plugged in until morning, so moving charging into this trough would deliver the same energy within the limit. A time-of-use (ToU) tariff such as Octopus Go, cheap from 00:30 to 05:30 [3], aims to do this. But every driver sees the same cheap window, so demand for its first half-hour jumps: on this day 11 chargers switched on together at 00:30,

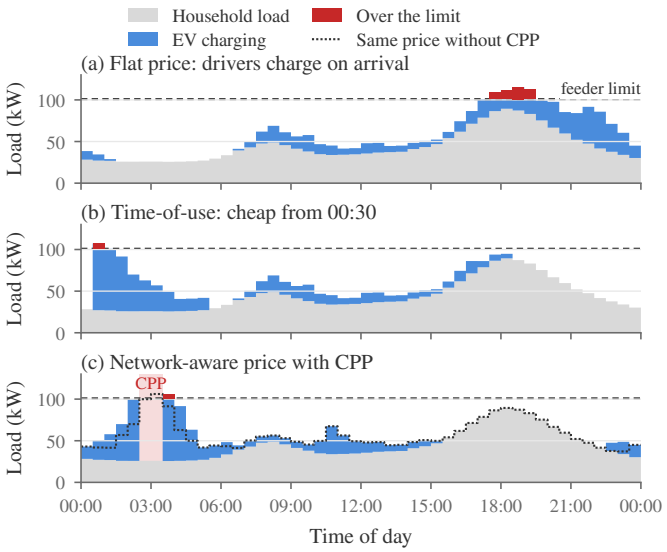


Fig. 1. Half-hourly load on one cold weekday on the studied London feeder, with EVs in 80% of homes and every charger responding to the price. (a) Uncontrolled charging adds to the evening peak. (b) A time-of-use tariff starts every charger at 00:30. (c) The network-aware price moves charging to around 03:00; its CPP empties 02:30–03:30, and the chargers all restart when it ends.

and the feeder went over its limit again (Fig. 1(b)). This is a *rebound peak*, a new peak created by the price signal itself.

Our earlier network-aware tariff avoids a single cheap window [1]. It prices each half-hour from the feeder's spare capacity, adds a critical-peak price (CPP), a surcharge high enough to stop charging, wherever forecast EV demand would use more than 80% of that capacity, and recomputes the price and the demand it creates until the two agree (fixed-point iteration). This relies on drivers responding differently. Wen *et al.* [4] found that about 20% of drivers respond strongly to charging prices, 60% respond slightly, and 20% hardly respond at all, so the demand that moves is spread over several half-hours.

Smart chargers remove this spread. New home chargers must now be smart [5]: a scheduling algorithm reads the next

day's prices and books the cheapest half-hours before the car is needed [6]. Every charger on the street runs a similar algorithm on the same price, so all of them choose the same half-hours. Trials already show the effect: network studies driven by charging data from 96 smart-charging homes on a ToU tariff found a new overnight peak that exceeds the evening peak once about 30% of homes have an EV [7], and data from 854 homes show a sharp spike at 00:30, when the cheap rate begins [8]. Fig. 1(c) shows three consequences for the network-aware tariff. *Demand for the cheapest half-hours rises*: without the CPP, the feeder reaches 106 kW at 03:00. *The CPP moves the peak rather than spreading it*: the next cheapest half-hour is the same for every charger, so they all wait for the surcharge to end and switch on in the same minute (11 chargers at 03:30). *The price never settles*: demand leaves whichever half-hour the CPP covers, so the CPP keeps following it. The mandated random start delay of up to 10 minutes [5] staggers switch-ons, but leaves the cars in the same half-hour.

Since every charger makes the same decision, a fix must either give the chargers different decisions or limit how many can act on one. We test three on the network-aware tariff: the random delay; a *price dither*, which adds a small household-specific offset of up to ± 2 p/kWh so that households have different cheapest half-hours; and a *quota*, which caps EV power in each half-hour and lets chargers book it in order of price. Our contributions are:

1) We show that when every charger is smart, a network-aware tariff makes the worst days worse than uncontrolled charging, its CPP never settles, and the CPP nearly doubles the chargers switching on together. Herding has been simulated for wholesale prices [9]; we show it for a network-aware tariff, minute by minute.
2) We test the random delay that chargers must already apply, which to our knowledge has not been done before, against a household price dither, adapted from the randomised group prices of [10], and a quota, similar to operating envelopes that limit each home's power draw [11].
3) We show that this holds for a commercial scheduler, and that mixing vendors' schedulers does not help a network-aware price, because it leaves no ties to break. All code and data are public[1].

II. Model

A. Demand and background load

Fig. 2 summarises the method. EV demand uses the session-level Monte Carlo model of [1]. Each day, a feeder with n_{EV} cars has $\mathrm{Poisson}(n_{\mathrm{EV}}k)$ home-charging sessions, where k is the average number of sessions per car per day. Each session draws a daily distance s, an arrival state of charge and a plug-in time, and requests $E = (1-\mathrm{SOC})\, s\, W_{100}/(100k)$, capped by the battery size. We add two things. First, each session has a departure time. Cars plugged in between 14:00 and 04:00 leave the next morning at $\mathcal{N}(07{:}30, 1\,\mathrm{h})$, and daytime plug-ins

[1]Code and data: https://github.com/RISKMASTER16/evalgopricing.

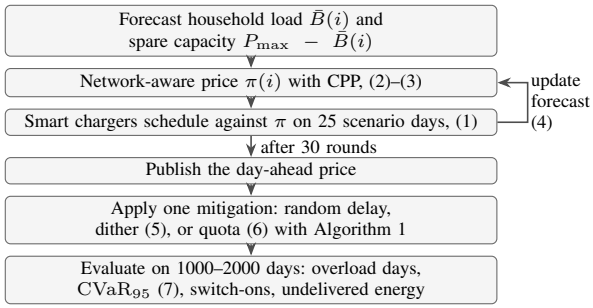


Fig. 2. The method. The price is recomputed against the charging it creates; each mitigation is then applied and evaluated on the same simulated days.

stay for 3–8 h. Requests are trimmed to what a 7.4 kW charger can deliver before departure, so any energy left undelivered is caused by the mitigation, not by the request. Second, the model runs in one-minute steps, because chargers that switch on together do so within minutes. Non-EV load is a half-hourly forecast profile scaled by a lognormal day factor with a coefficient of variation of 0.10, which is the operator's main forecast error.

B. Smart and legacy chargers

A share s of chargers is smart. A smart charger with plug-in time a, departure time b and price $\pi(t)$ solves

$$\min_{x} \sum_{t=a}^{b-1} \pi(t)\, x(t) \quad \text{s.t.} \quad \sum_{t} x(t)\, \eta\, \Delta t = E, \quad 0 \le x(t) \le P, \tag{1}$$

by charging at full power in the cheapest minutes. Prices are set per half-hour, so many minutes have the same price. A deterministic scheduler takes the earliest of these, and so does every other smart charger. Legacy chargers charge at full power as soon as they are plugged in.

Real chargers do not all run the same code, so we also model a *commercial* scheduler that follows the published logic of ev.energy [6], the platform used in the trial of [7]. It needs no price forecast, because day-ahead prices are published in advance. It adds an hour to the time the car needs, books that many half-hours, cheapest first and the later one when two cost the same, and charges through them in time order until full. In a *mixed* fleet, each charger uses one of the two schedulers at random, standing in for different vendors. Unless stated, results use (1).

C. Tariffs and mitigations

Shared prices. We compare four tariffs that every household sees identically: a flat 26.11 p/kWh tariff (uncontrolled charging); a static time-of-use (ToU) tariff at 7.5 p/kWh from 00:30 to 05:30 and 25 p/kWh otherwise; a wholesale-indexed tariff using the Agile formula [3] on a synthetic day-ahead price; and the network-aware tariff of [1], with and without its CPP. The network-aware price in half-hour i is

$$\pi(i) = C_1\, \bar{B}(i) + \mathrm{CPP}(i), \tag{2}$$

Algorithm 1 Hybrid quota: reservation in network-price order

Release capacity $C(t)$ for every minute t, from (6)
for each smart session, by plug-in time from 12:00 **do**
  Sort the minutes $t \in [a, b)$ by network price
  for each sorted minute t, until E is reserved **do**
    $x(t) \leftarrow \min\{P, C(t)\}$
    if $x(t) < P_{\min}$ **then** $x(t) \leftarrow 0$
    $C(t) \leftarrow C(t) - x(t)$
  end for
  Mark any energy not reserved as undelivered
end for

where $\bar{B}$ is the forecast non-EV load and C_1 sets the price at the household peak to 25 p/kWh, so charging is cheapest when the feeder is emptiest. The surcharge depends on how much of the spare capacity the forecast EV load $\hat{D}(i)$ would use, $u(i) = \hat{D}(i)/\big(P_{\max} - \bar{B}(i)\big)$:

$$\mathrm{CPP}(i) = \begin{cases} 0, & u(i) < 0.8, \\ 25\,\text{p/kWh}, & 0.8 \le u(i) < 1, \\ 200\,\text{p/kWh}, & u(i) \ge 1. \end{cases} \tag{3}$$

Because $\hat{D}$ depends on the price and the price depends on $\hat{D}$, the two are solved together by damped fixed-point iteration,

$$\hat{D}^{(m+1)} = (1-\alpha)\,\hat{D}^{(m)} + \alpha\, Q_{0.9}\Big[\mathcal{S}\big(\pi^{(m)}\big)\Big], \tag{4}$$

where $\mathcal{S}$ simulates the EV load that price $\pi^{(m)}$ creates on 25 scenario days using (1), $Q_{0.9}$ takes the 90th percentile across them so that the surcharge protects the bad days, and $\alpha = 0.35$. We run 30 rounds.

We then add one of three mitigations to the network-aware tariff.

Random delay. Each time a smart charger switches on, it waits $U(0, D_{\max})$, with $D_{\max} = 600$ s (the regulatory default) or 1800 s (the largest value an operator may set remotely) [5]. Each charging block keeps its length, so energy pushed past departure is lost.

Price dither. Household h sees

$$\pi_h(i) = \pi(i) + \delta\,\varepsilon_{h,i}, \qquad \varepsilon_{h,i} \sim U(-1, 1), \tag{5}$$

where $\varepsilon_{h,i}$ is drawn separately for each half-hour and redrawn daily, and $\delta = 2$ p/kWh. On average the tariff is unchanged; the offset only stops every charger from seeing the same cheapest half-hour.

Quota. The operator releases EV capacity

$$C(i) = P_{\max} - \bar{B}(i)\,Q_q(\xi) - F_{\text{leg}}(i) \tag{6}$$

in each half-hour, where $Q_q(\xi)$ is the q-quantile of the background day factor ($q = 0.9$ unless stated), so the release allows for a colder than expected day, and F_{leg} is the forecast load of legacy chargers. Smart chargers reserve capacity in the order they plug in, either as soon as possible (ASAP) or in order of the network price (the hybrid quota, Algorithm 1). A charger can run at reduced power, down to 1.4 kW.

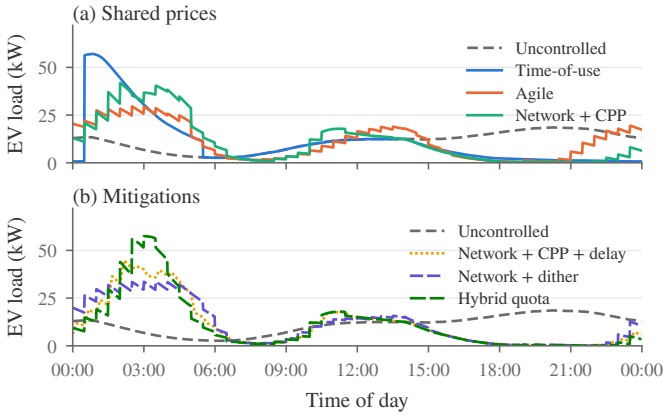


Fig. 3. Mean EV load over 2000 days, 80% EV ownership, all chargers smart. (a) Shared day-ahead prices. (b) Mitigations.

D. Metrics

For each simulated day d we take the peak $L_{\text{peak}}(d)$ of the half-hourly mean feeder load, since this is what heats cables and transformers. We report the share of days on which this peak exceeds the firm limit $P_{\max}$, and the conditional value at risk, the mean of the worst 5% of daily peaks [12],

$$\mathrm{CVaR}_{95} = \mathbb{E}\left[\, L_{\text{peak}} \mid L_{\text{peak}} \ge Q_{0.95}(L_{\text{peak}}) \right]. \tag{7}$$

The share of days over the limit shows how often the feeder is overloaded; CVaR_{95} shows how badly. To measure the rebound we count, each day, the largest number of chargers switching on in the same minute. For service we report the share of requested energy not delivered by departure.

III. Case Study

The feeder serves 75 homes and has a winter firm limit of 100.9 kW [1]. Background load follows a stylised winter weekday shape with a 72.0 kW peak and an overnight minimum near 21 kW. The mileage, state-of-charge and plug-in distributions come from [1], based on the National Travel Survey [13] and the DfT driver survey [14]. On average a car charges every 4.7 days and a session needs 19 kWh. The wholesale price is synthetic (daily mean 8 p/kWh, coefficient of variation 0.35, negative overnight prices on 18% of days). EV ownership runs from 20% to 100% of homes and the smart share from 0 to 1. Chargers draw 7.4 kW at 92% efficiency and can run down to 1.4 kW under a quota. The CPP uses tiers of 25 and 200 p/kWh at 80% and 100% of spare capacity.

Every tariff and mitigation is evaluated on the same simulated days (2000 days for Table I, 1000 for the sweeps). The CPP is solved on a separate set of days, so the evaluation is out of sample. Because the CPP does not settle when chargers are smart (Section IV-B), each CPP tariff is evaluated with the surcharge from each of the last 10 rounds, and we report the average.

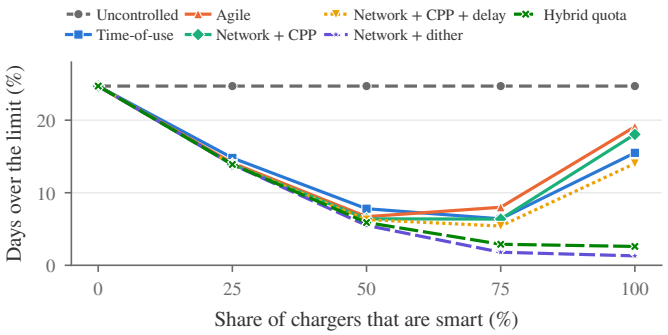


Fig. 4. Days over the firm limit against the share of smart chargers, 80% EV ownership, 1000 days.

IV. Results

A. Smart chargers crowd into the cheapest hours

Fig. 3(a) confirms that the day in Fig. 1 is typical: every shared price packs the evening charging into a few overnight half-hours. How much this matters depends on the share of smart chargers (Fig. 4). When half the chargers are smart, shared prices work well: the share of days over the limit falls from 24.7% to 6.4–7.8%, because smart and legacy chargers use different hours. As the smart share rises further, the smart chargers become the problem. With every charger smart, overload days rise back to 15.5–19.1%, and CVaR_{95} rises to 124–128 kW, above the 122 kW of uncontrolled charging. Shared prices then give fewer bad days than uncontrolled charging, but the bad days are worse.

B. The surcharge does not settle and causes rebound

With few smart chargers, the CPP fixed point of [1] settles. At 80% ownership and a smart share of up to 0.5, at most 1.3 surcharged slots changed per round over the last 10 rounds, and the EV forecast moved by at most 1.6 kW. With every charger smart, 3.0 slots changed per round and the forecast moved by 14.8 kW, rising to 4.7 slots and 17.4 kW at 100% ownership: the CPP keeps chasing the demand it moves. At 100% ownership the CPP left overload days unchanged (41.5% with it, 41.2% without) but nearly doubled the number of chargers switching on in the same minute, from 4.7 to 9.0 (Table I).

C. Mitigations

Random delay. Table I compares the mitigations. The 600 s delay cut synchronised switch-ons at 100% ownership from 9.0 to 2.7 chargers, as intended. But the cable heats with the load over the whole half-hour, and a few minutes' delay leaves the chargers in the same half-hour, so 36.3% of days still exceeded the limit (between 28.9% and 39.2% across the last 10 CPP rounds). Raising the delay to 1800 s did not help and left 1.45% of energy undelivered.

Price dither. A small offset is enough, because the cheapest half-hours of the network price differ by less than 0.5 p/kWh: a ±2 p/kWh offset changes which one is cheapest for each household. It cut overload days to 2.0% at 80% ownership and 7.0% at 100%, with all energy delivered (Fig. 3(b)). Adding the CPP back made the dither less effective on average (9.4%) and much less effective in some rounds (19.3%), because a 25 p/kWh surcharge outweighs a 2 p/kWh offset. Because chargers pick hours where their own offset is negative, the average price paid fell from 8.75 to 8.34 p/kWh; adding this difference to every price restores revenue without changing any schedule.

Quota. A cap on EV power keeps the worst days near the limit however the chargers behave, because no half-hour can be booked beyond its capacity. Booking in price order (the hybrid) gave 5.5% overload days and a CVaR_{95} of 102 kW at 100% ownership, against 8.6% and 107 kW when booking as soon as possible. The ASAP quota does worse because it fills the evening first, when headroom is smallest and a forecast error is most likely to cause an overload. The cost is service: the hybrid left 0.29% of energy undelivered.

D. Does the charger's algorithm matter?

Table II repeats the comparison with the commercial scheduler and the mixed fleet. The commercial scheduler herds just as much: every shared price exceeds the limit on 40.1–42.4% of days, against 37.8% for uncontrolled charging, and the CPP still raises the number of chargers switching on in the same minute from 3.5 to 8.9. A mixed fleet helps the ToU tariff (17.0%), because its long flat window lets the two schedulers

TABLE I
Results with All Chargers Smart, 2000 Days

Mechanism	80% ownership		100% ownership			
	Over (%)	CVaR_{95} (kW)	Over (%)	CVaR_{95} (kW)	Sync (ch.)	Undel. (%)
Uncontrolled	24.4	122	37.5	129	1.1	0.00
Time-of-use	15.8	125	38.1	147	9.5	0.00
Agile	18.6	127	41.5	150	8.5	0.00
Network, no CPP	18.1	128	41.2	151	4.7	0.00
Network + CPP	18.3	128	41.5	151	9.0	0.00
+ 600 s delay	14.7	124	36.3	146	2.7	0.38
+ 1800 s delay	14.7	124	36.7	147	1.8	1.45
+ dither	2.0	101	9.4	116	5.2	0.00
Network + dither	2.0	101	7.0	113	4.1	0.00
Quota (ASAP)	7.2	106	8.6	107	1.1	0.05
Hybrid quota	2.7	101	5.5	102	4.2	0.29

Over: days over the 100.9 kW limit. Sync: most chargers switching on in the same minute (daily mean). Undel.: energy not delivered by departure. Indented rows add a mitigation to Network + CPP.

TABLE II
Days Over the Limit (%) by Charger Scheduler, 100% Ownership, All Chargers Smart, 1000 Days

Price	Eq. (1)	Commercial	Mixed
Uncontrolled		37.8	
Time-of-use	40.0	41.6	17.0
Agile	44.0	40.1	36.3
Network, no CPP	43.4	42.1	38.2
Network + CPP	43.6	42.4	37.2
Network + dither ±2	6.3	13.3	9.0
Network + dither ±5	3.7	6.2	4.0

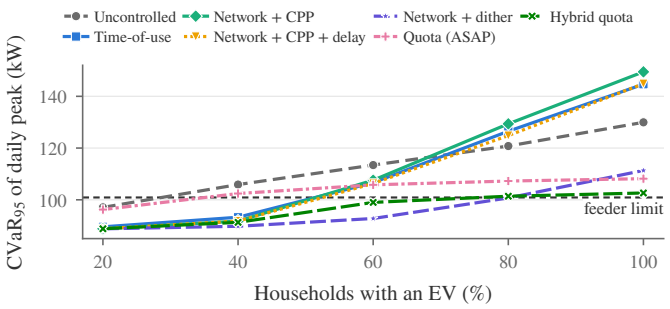


Fig. 5. CVaR_{95} of the daily peak against EV ownership, all chargers smart, 1000 days.

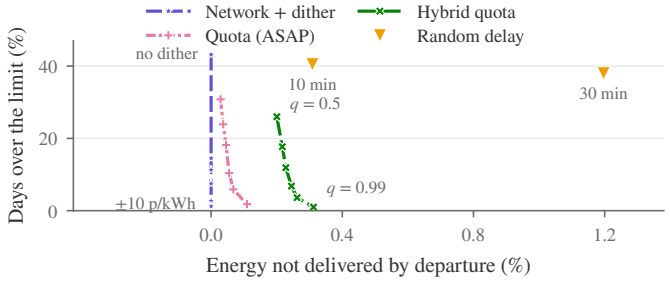


Fig. 6. Days over the limit against undelivered energy as each mitigation's setting varies, 100% EV ownership, all chargers smart, 1000 days. Quota confidence q runs from 0.5 (top) to 0.99 (bottom); δ is the dither half-width.

break ties at opposite ends. It barely helps the network price (37.2–38.2%), because every half-hour has a different price, so both schedulers choose the same ones. By removing ties, a network-aware price also removes the variety that different vendors would otherwise provide. The dither still works with the commercial scheduler, but needs to be larger: ± 2 p/kWh gives 13.3% and ± 5 p/kWh gives 6.2%.

E. Choosing a mitigation

Fig. 5 shows how the worst days grow with EV ownership. Shared prices beat uncontrolled charging up to 60% ownership and lose beyond it, reaching a CVaR_{95} of 145–149 kW at 100%, against 130 kW. The dithered network price stays within 1 kW of the firm limit up to 80% ownership, with no quota and all energy delivered. Beyond that, only the hybrid quota stays near the limit (103 kW at 100%).

Fig. 6 shows each mitigation's trade-off at 100% ownership. A larger dither lowers overload days with no loss of service: 2.7% at ± 5 p/kWh and 1.2% at ± 10 p/kWh, at a revenue cost of 1.42 p/kWh at ± 5 p/kWh. A more cautious quota (q from 0.5 to 0.99) cuts the hybrid's overload days from 26.0% to 1.0%, while undelivered energy rises from 0.20% to 0.31%. The delay barely helps at any setting.

V. Conclusions

Smart chargers all read the same price and run similar algorithms, so they all make the same decision. On the studied London feeder with an EV in every home, this made every shared price, including the network-aware tariff, worse than uncontrolled charging on the worst days, stopped the CPP from settling, and nearly doubled the number of chargers switching on together, also with a commercial scheduler. The mandated random delay removes the minute-scale spike but not the half-hourly peak.

Two changes fix this. A slightly different price for each household gives the chargers different cheapest half-hours, and kept the worst days at the limit up to 80% EV ownership with all energy delivered. Above that, only a quota filled in network-price order kept the worst days near the limit, leaving under 0.3% of energy undelivered. A supplier can apply a dither to chargers already installed; a quota needs the network operator to tell each charger how much power it may use. This method can be applied to other feeders by substituting the relevant local parameters, such as travel data, household demand and feeder ratings, giving network operators a practical way to test a tariff against smart chargers before it is introduced.

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